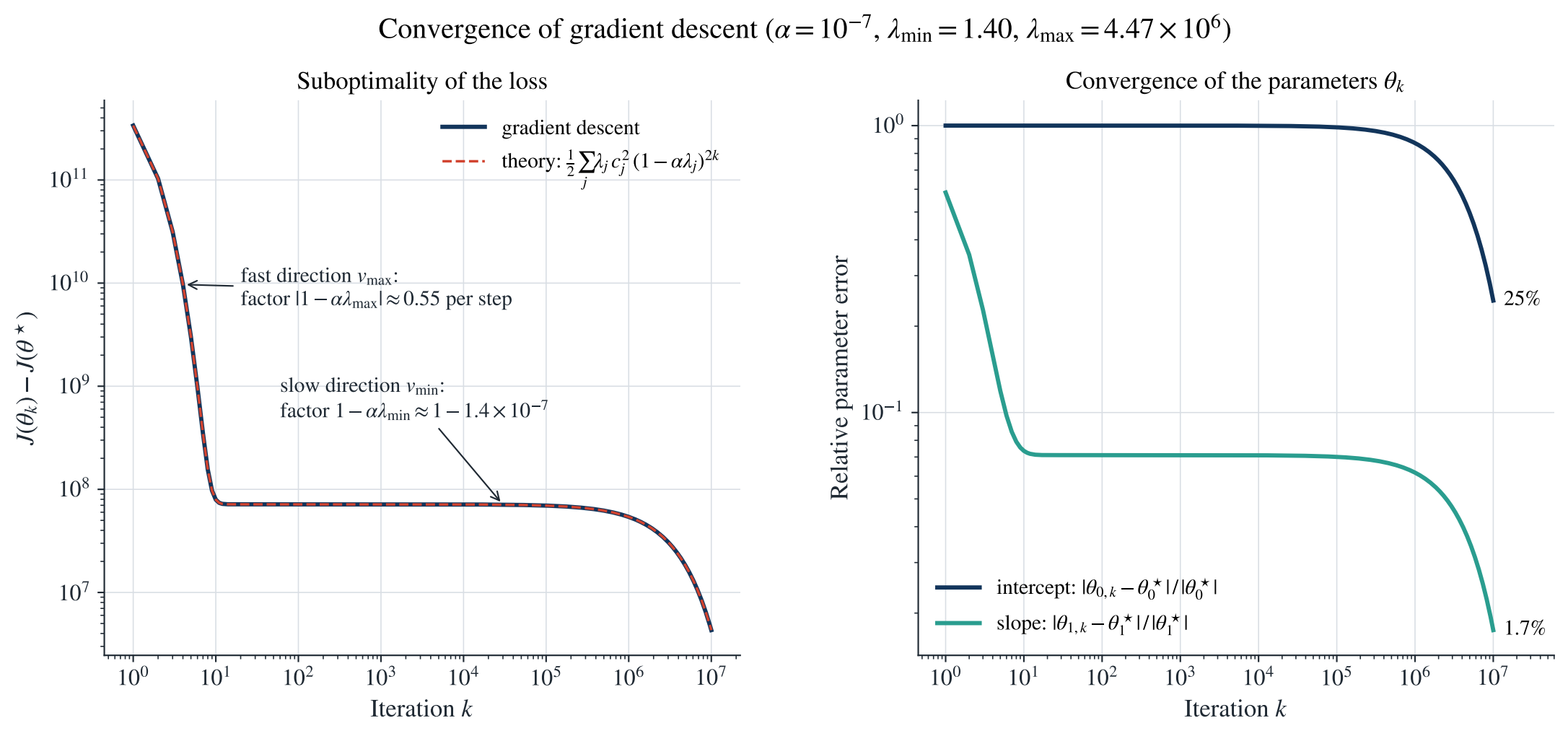
Convergence of gradient descent ($\alpha = 10^{-7}$, $\lambda_{\min} = 1.40$, $\lambda_{\max} = 4.47 \times 10^{6}$)
Suboptimality of the loss
gradient descent
theory: $\frac{1}{2}\sum_j \lambda_j c_j^2 (1-\alpha\lambda_j)^{2k}$
fast direction $v_{\max}$:
factor $|1-\alpha\lambda_{\max}| \approx 0.55$ per step
slow direction $v_{\min}$:
factor $1-\alpha\lambda_{\min} \approx 1 - 1.4\times10^{-7}$
$J(\theta_k) - J(\theta^\star)$
Iteration k
Convergence of the parameters θ_k
Relative parameter error
intercept: $|\theta_{0,k} - \theta_0^\star| / |\theta_0^\star|$
slope: $|\theta_{1,k} - \theta_1^\star| / |\theta_1^\star|$
25%
1.7%
Iteration k